

index $i \sim N^a$ we have a transition (from strong dependence to independence) on a scale $k \sim N^{2(1-a)/3}$. The situation in the bulk, $a = 1$, is very different and will be explained separately at the end (see also Figure 2).

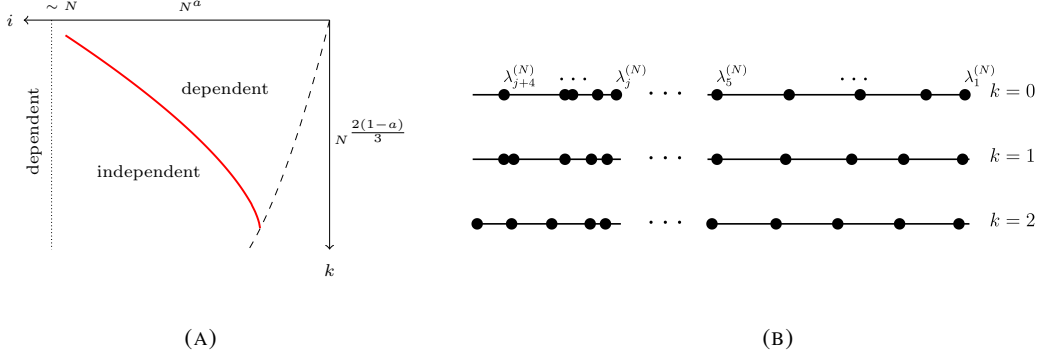


FIGURE 2. (A): Decorrelation transition in the intermediate regime $a \in [0, 1)$. The figure depicts the shrinking spectrum of the minors $H^{(N-k)}$; the dashed line their largest eigenvalue. We consider $a \in [0, 1)$, fix an index $i \sim N^a$ and track the trajectory of $\lambda_i^{(N-k)}$ as k decreases. This trajectory crosses the solid red line at the level $k(a) \sim N^{2(1-a)/3}$, marking the phase transition: for $k \ll k(a)$, $\lambda_i^{(N-k)}$ and $\lambda_i^{(N)}$ remain highly correlated, while for $k \gg k(a)$, $\lambda_i^{(N-k)}$ becomes essentially independent of $\lambda_i^{(N)}$. For $a = 1$, i.e. to the left of the dotted line (bulk regime), the eigenvalues remain correlated for any k .

(B): Individual eigenvalues of $H^{(N)}$ and its first two minors in the extreme edge regime and in the bulk regime. In the edge regime, the i -th eigenvalues $\lambda_i^{(N)}, \lambda_i^{(N-1)}, \lambda_i^{(N-2)} \dots$ stick to each other indicating very strong correlation. In the bulk regime the only correlation present is the one enforced by interlacing.

We now explain how to obtain the threshold $k \sim N^{2(1-a)/3}$ for $a \in [0, 1)$. We do not present the mathematical details for brevity, but just explain the general strategy. We separate the argument into two parts: we first show that the eigenvalues corresponding to indices $i \sim N^a$ are (asymptotically) independent for $k \gg N^{2(1-a)/3}$ and then that they are strongly correlated for $k \ll N^{2(1-a)/3}$.

For $k \gg N^{2(1-a)/3}$, we follow a strategy similar to Section 3, i.e. we show the independence of the eigenvalues using a Dyson Brownian motion (DBM) argument. More precisely, we couple the time evolution of $\lambda_i^{(N)}(t)$ and $\lambda_i^{(N-k)}(t)$ under a Brownian matrix flow with two fully independent DBM flows $\mu_i^{(1)}(t), \mu_i^{(2)}(t)$ showing that for larger times t they are close to each other. To control the coupling, technically this requires following analogue of the eigenvector overlap bound (cf. (3.1)):

$$(2.8) \quad |\langle \mathbf{w}_i^{(N)}, \mathbf{w}_j^{(N-k)} \rangle|^2 \prec \frac{N^{2(1-a)/3}}{k} \wedge 1, \quad i, j \sim N^a.$$

The proof of (2.8) is very similar to the one in Section 4.2 below, once it is taken into consideration that the local eigenvalue density is of order $\sim N^{(1-a)/3}$, when $i \sim N^a$. The bound (2.8) shows that the eigenvectors of $H^{(N)}$ and $H^{(N-k)}$ are asymptotically orthogonal $|\langle \mathbf{w}_i^{(N)}, \mathbf{w}_j^{(N-k)} \rangle| \ll 1$ for $k \gg N^{2(1-a)/3}$. Armed with this bound as an input, we can then proceed with a DBM argument similar to the one in Section 5. The only difference is that in the current case we cannot directly rely on the results from [14, 53], which are formulated only in the regime $i \sim 1$. However, inspecting their proof, it is clear that similar arguments can be extended to the regime $i \sim N^a$, for $a \in (0, 1)$. This gives the desired independence when $k \gg N^{2(1-a)/3}$.

For $k \ll N^{2(1-a)/3}$, we can follow a strategy similar to Section 7. As the level repulsion scale for $\lambda_i^{(n)} - \lambda_{i+1}^{(n)}$, for $n \in [N-k, N]$, should now be changed to $n^{-(2+a)/3-\delta}$, the error term in each step of the recursion analogous to (7.11) will be $N^{(a-1)/3+C\delta}$ instead of $N^{-1/3+C\delta}$. Hence, after k steps of the